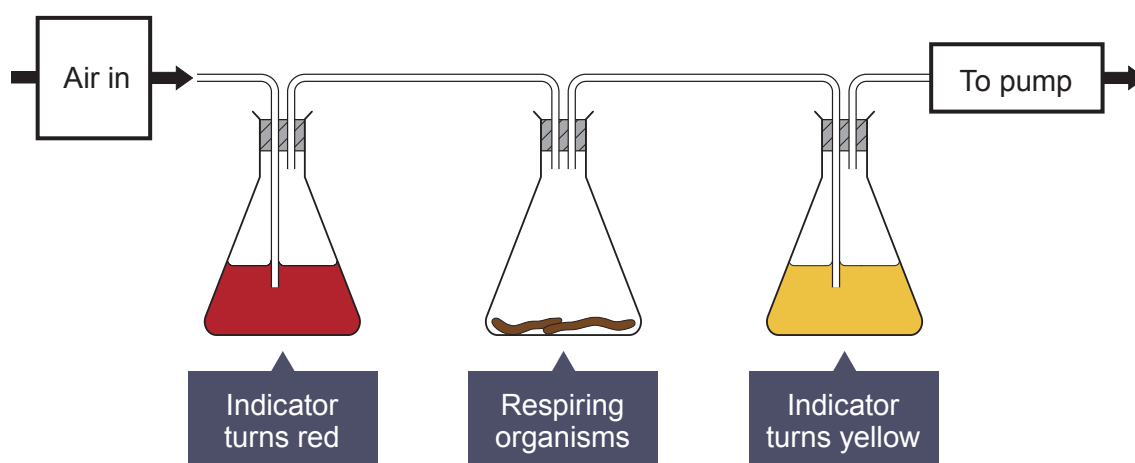


7. Hydrogen carbonate indicator can be used to measure the rate of respiration.

Concentration of carbon dioxide (%)	Colour of hydrogen carbonate indicator
less than 0.04	purple
0.04 (normal concentration in air)	red
greater than 0.04	yellow

The apparatus below can be used to compare the rate of respiration in small animals.



A group of students used different respiring organisms and recorded the time taken for the hydrogen carbonate indicator to turn yellow. The apparatus was kept at a constant temperature of 20 °C.

The results are shown in the table below.

Organism	Time taken for the hydrogen carbonate indicator to turn yellow (min)
earthworm	22
woodlouse	18
grasshopper	10
mouse	3

- (a) Explain why the hydrogen carbonate indicator changed colour in this experiment. [2]

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- (b) Give **two** reasons why the hydrogen carbonate indicator turned yellow fastest with the mouse. [2]

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- (c) The experiment was performed at a temperature of 20 °C. [2]

State **two** other variables that need to be controlled when comparing the respiration rates of different organisms.

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- (d) Describe how you would set up a control for this experiment and state the expected result. [3]

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END OF PAPER

THE PERIODIC TABLE

Group

1 H Hydrogen 1																				4 He Helium 2															
7	Li	9	Be											11	B	12	C	14	N	16	O	19	F	20	Ne										
	Lithium		Beryllium												Boron		Carbon		Nitrogen		Oxygen		Fluorine		Neon										
23	Na	24	Mg											27	Al	28	Si	31	P	32	S	35.5	Cl	40	Ar										
	Sodium		Magnesium												Aluminium		Silicon		Phosphorus		Sulfur		Chlorine		Argon										
39	K	40	Ca	45	Sc	48	Ti	51	V	52	Cr	55	Mn	56	Fe	59	Co	59	Ni	63.5	Cu	65	Zn	70	Ga	73	Ge	75	As	79	Se	80	Br	84	Kr
	Potassium		Calcium		Scandium		Titanium		Vanadium		Chromium		Manganese		Iron		Cobalt		Nickel		Copper		Zinc		Gallium		Germanium		Arsenic		Selenium		Bromine		Krypton
86	Rb	88	Sr	89	Y	91	Zr	93	Nb	96	Mo	99	Tc	101	Ru	106	Rh	108	Pd	112	Ag	115	Cd	119	In	122	Sn	127	Sb	128	Te	131	Xe		
	Rubidium		Strontium		Yttrium		Zirconium		Niobium		Molybdenum		Technetium		Ruthenium		Rhodium		Palladium		Silver		Cadmium		Tin		Antimony		Tellurium		Iodine		Xenon		
133	Cs	137	Ba	139	La	179	Hf	181	Ta	184	W	186	Re	190	Os	195	Ir	197	Pt	201	Au	204	Hg	207	Pb	209	Bi	210	Po	210	At	222	Rn		
	Caesium		Barium		Lanthanum		Hafnium		Tantalum		Tungsten		Rhenium		Osmium		Iridium		Platinum		Gold		Mercury		Lead		Bismuth		Polonium		Astatine		Radon		
223	Fr	226	Ra	227	Ac																														
	Francium		Radium		Actinium																														

Key

relative atomic mass

Name	Symbol	A_r

atomic number